Satellite and Microscale Geographic Prediction of PM2.5 Across Delhi Under Station-Level Spatial Validation

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Abstract--Fine-scale PM2.5 information is needed for urban exposure assessment, but fixed monitors are too sparse to describe local gradients across Delhi. This paper documents a station-validated PM2.5 modeling pipeline that joins Central Pollution Control Board (CPCB) 10 AM observations with CPCB station coordinates, Landsat TOA reflectance, spectral indices, AOD, GEOS-CF meteorology, SRTM terrain, and OpenStreetMap land-use features. The final all-features table contains 3,289 complete station-date observations from 31 stations during 2021-2025 and 130 predictors. Multiple linear regression, GTWR-style local weighted regression, and LightGBM were compared under validation designs that separate random reference splits from spatial tests. The primary test is leave-one-station-out cross-validation because it holds out complete stations and avoids repeated-station leakage. LightGBM_d5 achieved R^2 = 0.775, RMSE = 39.75 ug/m3, MAE = 26.90 ug/m3, and near-zero bias. A station-cluster bootstrap gave a 95% R^2 interval of 0.714 to 0.822, and paired station-level tests favored LightGBM_d5 over GTWR. Residual bias showed no spatial autocorrelation under 5-nearest-neighbor Moran's I.

Index Terms--Air pollution, Delhi, LightGBM, PM2.5, spatial cross-validation, station-LOSO, TOA-LUR, urban exposure modeling.

I. INTRODUCTION

PM2.5 exposure is a major public-health concern in Indian cities, but the measurement infrastructure available for spatially resolved analysis remains sparse. Fixed monitors are reliable at their own locations; they do not, by themselves, describe the large variation produced by roads, land use, local sources, vegetation, atmospheric transport, and seasonal emissions. For health analysis and route planning, the required object is not only a daily city mean but a reproducible surface that can estimate concentrations at locations with no monitor.

The modeling strategy used here follows the practical idea behind top-of-atmosphere land-use-regression studies: combine remotely sensed optical and aerosol information with local geographic descriptors that act as source, dispersion, and land-cover proxies. In Delhi, this means joining Landsat TOA reflectance and spectral indices, AOD, meteorology, terrain, and OpenStreetMap features with ground PM2.5 observations. The goal is not to claim that every street segment has already been independently validated. The first goal is to establish whether a station-date model can generalize to unseen monitoring locations without spatial leakage.

Delhi is a useful test case because it has severe PM2.5 pollution, strong seasonal variation, diverse source environments, and enough monitoring stations to evaluate station-level spatial generalization. The 2021-2025 period also contains multiple seasonal cycles, allowing the model to learn winter and postmonsoon high-pollution behavior as well as lower-variance monsoon behavior. These conditions make it possible to test whether the model is learning a transferable spatial-environmental relationship rather than memorizing station identity.

The main methodological constraint is validation. Random record-level splits are not sufficient because many rows come from the same station on different dates. A random split can therefore place nearly identical station context in both training and evaluation. This paper treats leave-one-station-out cross-validation as the principal test. Each fold holds out all dates from one station, so the model must predict a location whose station identity was not seen during training.

The paper contributes a CPCB-only Delhi feature pipeline, a no-leakage station-LOSO validation protocol, and diagnostics for model choice, seasonal behavior, buffer-radius sensitivity, and difficult stations.

II. DATA AND FEATURE ENGINEERING

A. Study Area and Monitoring Data

The study area is Delhi, India. The monitoring network contains 31 Central Pollution Control Board (CPCB) stations with station coordinates. The target variable is hourly CPCB PM2.5 reduced to a single 10 AM local observation per station-date. The 10 AM target is intentional: it aligns the ground observation with the Landsat overpass window more closely than daily mean PM2.5 and reduces temporal mismatch between surface observations and satellite predictors. The station inventory covers 2021-01-01 to 2025-12-31.

The raw CPCB files were stored as station and year-specific CSVs. The cleaning script parsed all available Delhi files, normalized date-time fields, extracted PM2.5 at 10 AM local time, and attached station metadata. Stations that could not be matched to reliable CPCB station coordinates were removed before modeling. This step produced a larger target inventory than the final modeling table; the final row count is smaller because every row also needs complete Landsat, AOD, meteorology, terrain, and OSM predictors.

TABLE I
STATION NETWORK AND MODELING TABLE SUMMARY

Item	Value
Stations	31
Study period	2021-01-01 to 2025-12-31
Valid 10 AM station records	52,957 of 56,606 possible
Mean station missingness	6.45%
Complete all-feature modeling rows	3,289
Feature count in main model	130

B. Google Earth Engine Extraction

Google Earth Engine was used as a set of separate extraction modules rather than as one undifferentiated data source. The matched CPCB station GeoJSON was rendered into a reusable station block, and that same station geometry was pasted into each GEE script. This kept station identifiers and coordinates consistent across Landsat, meteorology, AOD, and terrain exports. Each module exported a station-date or station-level CSV to Google Drive, and the Python feature-engineering scripts merged those CSVs later by station and date.

Landsat module. The Landsat extraction produced the optical surface/atmospheric predictors used by the TOA-LUR feature set. For each station-date, the script exported top-of-atmosphere bands B2, B3, B4, and B7 and calculated NDVI, EVI, and IBI at 30 m scale over 500 m station buffers. The Landsat export also retained solar azimuth and solar zenith. These two geometry fields were not treated as metadata only; they were kept as predictors because acquisition geometry and seasonal illumination are strongly aligned with Delhi's PM2.5 seasonal cycle.

Meteorology module. The meteorology script exported GEOS-CF fields at the same station coordinates and dates, using the 05:00 UTC image to approximate the 10:30 AM IST Landsat overpass and a 25 km extraction scale. The retained variables were wind speed, temperature, sea-level pressure, and relative humidity. These variables represent dispersion and boundary-condition proxies: stagnant, cool, high-pressure conditions are expected to coincide with higher PM2.5, while wind speed and humidity help explain day-to-day concentration changes that cannot be inferred from land use alone.

AOD module. The AOD script was run separately because aerosol retrieval availability differs from Landsat and meteorology availability. The MODIS MCD19A2 MAIAC product was sampled at 1 km scale. The merged model table uses filled 550 nm AOD, filled 470 nm AOD, filled AOD uncertainty, and a binary daily-availability flag. The flag is kept because a filled AOD value and an observed same-day AOD value do not carry the same information. This lets the model learn both aerosol loading and whether the daily satellite aerosol retrieval was actually available.

Terrain module. Terrain was exported as a static station-level table from SRTM-derived elevation, slope, and aspect at 30 m scale. These fields do not vary by date, so they are joined by station identifier rather than by station-date. Keeping terrain in its own export also prevents accidental duplication of static fields during the daily feature merge.

C. OSM Land-Use and Microscale Features

OpenStreetMap features were calculated around each station at 50, 200, 500, 1000, and 2000 m radii. The implementation uses projected geometry for area and length calculations rather than raw WGS84 coordinates, so these are buffer-scale vector features rather than raster pixels. Feature families include road counts and densities, road-type densities, building counts and building densities, restaurant, temple, and bus-stop counts, and land-use fractions. This produced 110 OSM features, with 22 features at each buffer radius. The multiple radii are retained because microscale traffic indicators and neighborhood land-use context operate at different spatial scales.

D. Feature Table Assembly

The final modeling table is a station-date panel. CPCB 10 AM PM2.5 defines the target; Landsat, AOD, and meteorology define date-varying predictors; terrain and OSM features define station-specific spatial predictors. The join keys are station identifier and date for dynamic features, and station identifier for static features. Rows with missing required predictors were dropped for the all-features experiment, yielding 3,289 complete station-date observations from 31 stations.

The all-features model used 130 predictors. A smaller core feature set contains Landsat TOA bands, spectral indices, solar geometry, meteorology, and terrain. The core+AOD set adds the four AOD variables. The full set then appends the 110 OSM buffer variables. This staged design is useful because it allows the paper to separate the effect of satellite and meteorological predictors from the incremental value of local land-use descriptors.

III. MODELING AND VALIDATION PROTOCOL

A. Model Families

Three model families were evaluated: multiple linear regression with standardized predictors, GTWR-style local weighted linear regression with ridge regularization, and LightGBM gradient-boosted decision trees. MLR is the transparent linear baseline. GTWR is included because it is a common spatiotemporal local-regression approach for air-pollution surfaces. LightGBM is included because it can learn nonlinear interactions among satellite, meteorological, seasonal, and OSM predictors while still working with tabular features.

GTWR used a reduced 42-feature subset rather than the complete 130-feature table. This was a stability decision: local weighted regressions can become ill-conditioned when the number of features is large relative to the number of effective nearby observations in a spatial-temporal kernel. The GTWR variants therefore function as disciplined local-regression baselines, not as exhaustive hyperparameter searches.

B. LightGBM Configuration

The main LightGBM model was the depth-5 configuration shown in Table II. The selected settings use slow learning, moderate tree depth, column and row subsampling, and a fixed random seed so that the comparison can be reproduced.

TABLE II
LIGHTGBM_D5 HYPERPARAMETER CONFIGURATION

Parameter	Value
objective	regression
n_estimators	500
learning_rate	0.02
max_depth	5
num_leaves	31
min_child_samples	15
subsample	0.9
colsample_bytree	0.9
random_state	42
verbose	-1

A shallower LightGBM_d3 model was also evaluated to test whether reducing tree depth improved temporal or station generalization. The selected main model is the depth-5 configuration because it gives the best station-LOSO performance in the full all-features comparison.

C. Split Strategy

The main validation design is leave-one-station-out cross-validation. For each fold, all observations from one station are held out, the model is trained on the remaining stations, and predictions are made for the held-out station. This is the primary estimate of spatial generalization because random K-fold validation can place records from the same station in both training and test sets.

Additional splits are reported for interpretation. Random 5-fold validation is kept as a reference but is not used as the headline performance claim. Date group validation holds out dates across stations and tests temporal episodes. Station-group validation holds out groups of stations and approximates a less granular spatial holdout. Year-LOSO holds out complete years and is the most direct test of temporal extrapolation. Reporting these side by side prevents the paper from mixing an optimistic record-level score with a stricter spatial claim.

TABLE III
LIGHTGBM_D5 R^2 AND RMSE BY SPLIT ROLE

Role	Split	n	R^2	RMSE
Train, in-sample	train_full_data	3,289	0.915	24.51
Random validation	random_5fold	3,289	0.821	35.46
Date holdout validation	date_group_5fold	3,289	0.536	57.13
Station-group validation	station_group_5fold	3,289	0.777	39.62
Station-LOSO spatial test	station_loso	3,289	0.775	39.75
Year-LOSO temporal test	year_loso	3,289	0.573	54.78

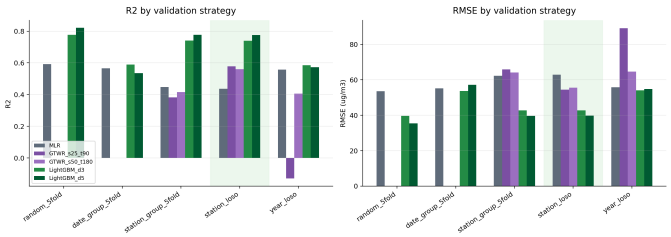


Fig. 1. Model R^2 and RMSE by validation split.

D. Statistical Testing and Diagnostics

Station heterogeneity was handled with a station-cluster bootstrap. Stations were resampled with replacement, and all held-out predictions from sampled stations were included in each draw. Paired model comparisons used station-level Delta R^2 and a Wilcoxon signed-rank test across the 31 stations. Residual spatial autocorrelation was tested using Moran's I on station-level mean bias with symmetric 5-nearest-neighbor weights.

All metrics are calculated on predictions from the relevant held-out split. R^2 is reported together with RMSE, MAE, and bias because no single metric is sufficient for air-pollution modeling. R^2 describes explained variance, RMSE penalizes high-pollution misses, MAE is more robust to extreme days, and bias tests whether the model systematically overpredicts or underpredicts a station or season.

IV. RESULTS

A. Station-LOSO Model Comparison

The all-features LightGBM_d5 model had the strongest station-LOSO performance. It achieved R^2 = 0.775, RMSE = 39.75 ug/m3, MAE = 26.90 ug/m3, and bias = 0.20 ug/m3 across 3,289 held-out predictions.

TABLE IV
STATION-LOSO MODEL PERFORMANCE

Model	n	R^2	RMSE	MAE	Bias
LightGBM_d5	3,289	0.775	39.75	26.90	0.20
LightGBM_d3	3,289	0.740	42.80	29.88	0.19
GTWR_s25_t90	3,289	0.578	54.44	40.93	-0.75
GTWR_s50_t180	3,289	0.560	55.62	42.00	-0.46
MLR	3,289	0.436	62.95	46.09	-4.49

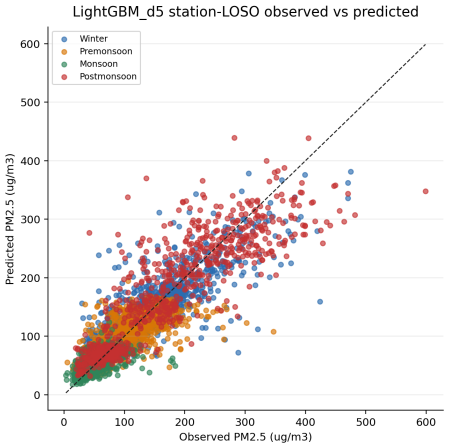


Fig. 2. Observed and predicted PM2.5 under station-LOSO.

B. Uncertainty and Paired Model Tests

The station-cluster bootstrap showed that the main station-LOSO result is stable across stations. LightGBM_d5 retained a clear advantage over the GTWR and MLR baselines after uncertainty was calculated at the station level.

TABLE V
BOOTSTRAP R^2 INTERVALS

Model	Estimate	CI low	CI high
LightGBM_d5	0.775	0.714	0.822
GTWR_s25_t90	0.578	0.518	0.628
GTWR_s50_t180	0.560	0.509	0.601
MLR	0.436	0.273	0.547

TABLE VI
PAIRED STATION-LEVEL WILCOXON TESTS

Comparison	Mean Delta	Median Delta	p
LightGBM_d5 > GTWR_s25_t90	0.198	0.192	1.16e-08
LightGBM_d5 > GTWR_s50_t180	0.220	0.206	4.10e-08
LightGBM_d5 > MLR	0.405	0.223	2.33e-09

The paired tests support the model choice. LightGBM_d5 was not only better in aggregate; it was better in the paired station-level comparison against GTWR and MLR.

C. Seasonal and Temporal Generalization

Seasonal performance varied. The model was strongest in postmonsoon and winter and weakest in monsoon. The lower monsoon R^2 is consistent with a compressed PM2.5 concentration range, even though absolute error is also lower.

TABLE VII
LIGHTGBM_D5 STATION-LOSO PERFORMANCE BY SEASON

Season	n	R^2	RMSE	MAE	Bias
Winter	871	0.639	44.98	31.53	0.14
Premonsoon	1,014	0.516	32.84	23.31	-0.79
Monsoon	627	0.331	21.24	14.93	1.65
Postmonsoon	777	0.764	51.48	36.07	0.39

TABLE VIII
LIGHTGBM_D5 YEAR-LOSO PERFORMANCE BY HELD-OUT YEAR

Year	n	R^2	RMSE	MAE	Bias
2021	536	0.569	54.99	40.34	-4.57
2022	809	0.536	50.18	35.88	-1.47
2023	584	0.650	51.53	36.07	-8.53
2024	727	0.555	52.89	40.76	-0.90
2025	633	0.554	64.42	45.36	2.57

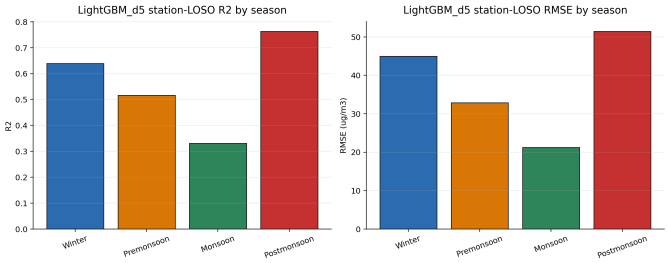


Fig. 3. Seasonal station-LOSO R^2 and RMSE.

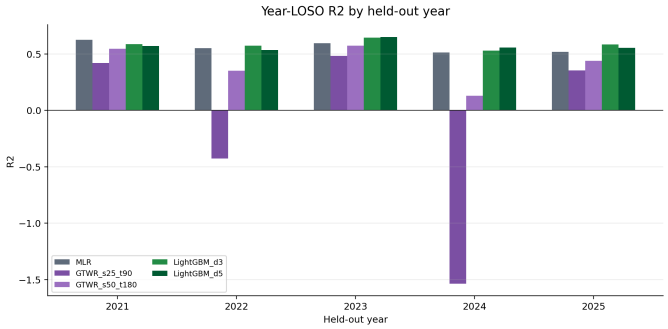


Fig. 4. Held-out-year temporal generalization.

Year-LOSO performance was weaker than station-LOSO. The aggregate LightGBM_d5 year-LOSO R^2 was 0.573 with RMSE = 54.78 ug/m3, showing that temporal extrapolation is harder than spatial prediction across the current station network.

D. Feature Interpretation

The feature correlations and feature-importance artifacts should be interpreted as complementary diagnostics. Pearson correlation measures marginal linear association with PM2.5 and therefore highlights seasonally aligned variables such as solar geometry, pressure, AOD, reflectance, vegetation indices, and temperature. LightGBM feature importance can then capture nonlinear and interaction effects, including cases where OSM variables are weak marginally but useful when combined with season and meteorology.

TABLE IX
TOP ABSOLUTE PEARSON CORRELATIONS WITH PM2.5

Feature	Meaning	Family	Corr.
sun_zenith	Solar angle season proxy	solar_geometry	0.529
SLP_kpa	Pressure and stagnation proxy	meteorology	0.511
sun_azimuth	Solar direction season proxy	solar_geometry	0.469
AOD_470_filled	Shortwave aerosol loading proxy	aod	0.459
AOD_550_filled	Visible aerosol loading proxy	aod	0.455
NDVI	Vegetation greenness spectral index	spectral_indices	-0.450
B2	Blue reflectance haze signal	landsat_toa	0.447
T_celsius	Temperature and season signal	meteorology	-0.434
EVI	Enhanced vegetation greenness index	spectral_indices	-0.336
AOD_daily_available	Observed aerosol retrieval flag	aod	0.248
WS_ms	Wind speed dispersion proxy	meteorology	-0.241
B3	Green reflectance haze signal	landsat_toa	0.235

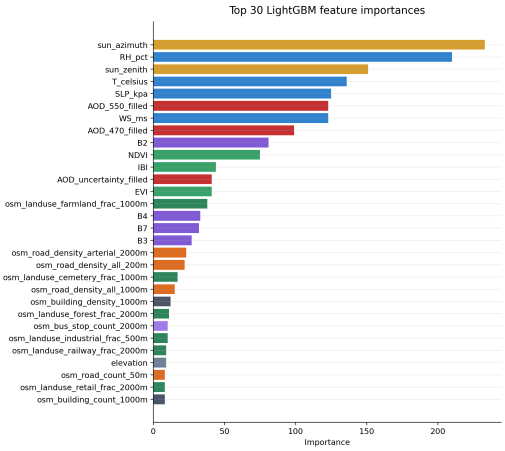


Fig. 5. Top LightGBM feature importances.

The strongest positive correlations are sun zenith, sea-level pressure, sun azimuth, AOD at 470 and 550 nm, and blue-band reflectance. The strongest negative correlations include NDVI, temperature, EVI, wind speed, AOD uncertainty, and SWIR reflectance. This pattern is consistent with Delhi's seasonal cycle: high PM2.5 days tend to occur under cooler, more stagnant, higher-pressure conditions with lower vegetation indices and higher aerosol loading.

E. Buffer-Radius Sensitivity

The OSM feature design includes five buffer radii, so the paper must justify whether all are needed. The buffer sensitivity experiment retrained the same LightGBM_d5 protocol with core variables, core+AOD variables, single-radius OSM variants, and cumulative OSM variants. The best single radius was 1000 m (R^2 = 0.781, RMSE = 39.21 ug/m3). The all-radii model was very close (R^2 = 0.775, RMSE = 39.75 ug/m3) and retained richer multiscale context. This supports keeping the multiradius representation while noting that 1000 m is the strongest single-radius scale in the current station network.

TABLE X
BUFFER-RADIUS SENSITIVITY UNDER STATION-LOSO

Variant	Feat.	OSM	R^2	RMSE
Core only	16	0	0.768	40.43
Core + AOD	20	0	0.765	40.61
All radii	130	110	0.775	39.75
OSM 50 m only	42	22	0.765	40.62
OSM 200 m only	42	22	0.759	41.13
OSM 500 m only	42	22	0.767	40.43
OSM 1000 m only	42	22	0.781	39.21
OSM 2000 m only	42	22	0.775	39.81

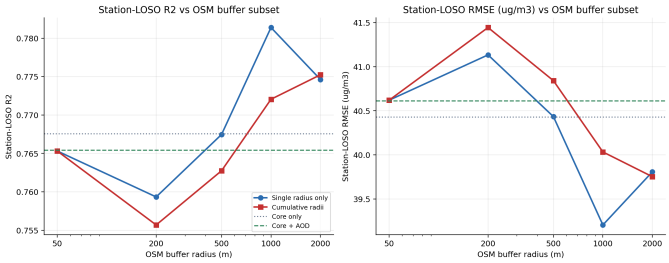


Fig. 6. Buffer-radius sensitivity under station-LOSO.

F. Residual Spatial Autocorrelation

Station-level residual bias from LightGBM_d5 did not show spatial autocorrelation. Moran's I was -0.009 with p = 0.927 using symmetric 5-nearest-neighbor weights. This suggests that the model did not leave a broad spatially clustered bias pattern across the station network.

G. High-Error Station Diagnostic

The worst station-level R^2 for LightGBM_d5 occurred at Shadipur (CPCB_8179). The station had R^2 = 0.037, RMSE = 62.24 ug/m3, MAE = 46.39 ug/m3, and bias = 18.04 ug/m3. The postmonsoon period was especially difficult, with negative R^2 and high RMSE.

TABLE XI
SHADIPUR HELD-OUT STATION PERFORMANCE

Model	R^2	RMSE	MAE	Bias
GTWR_s50_t180	0.224	55.88	43.44	2.36
MLR	0.160	58.12	45.90	12.03
GTWR_s25_t90	0.157	58.24	45.20	2.90
LightGBM_d3	0.150	58.48	44.64	17.20
LightGBM_d5	0.037	62.24	46.39	18.04

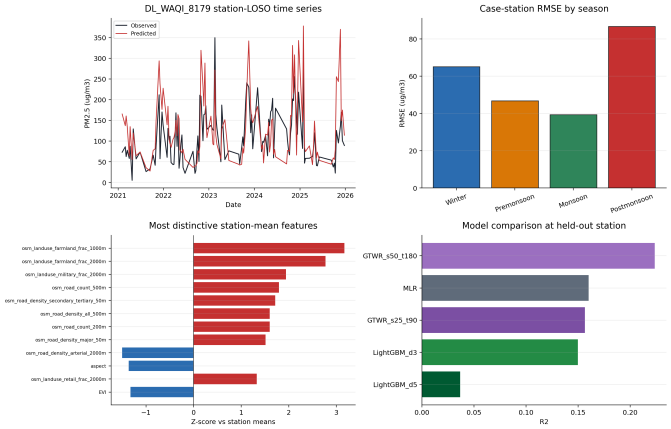


Fig. 7. Shadipur high-error station diagnostic.

Shadipur has distinctive station-mean features compared with the rest of the network, including high farmland fraction within 1-2 km, high road count and road density, low vegetation indices, and an unusual land-use mix. This case is useful because it shows where a high aggregate score still needs local source diagnostics.

V. DISCUSSION

The main result is that a satellite and microscale geographic feature set can predict Delhi PM2.5 at unseen station locations with strong performance. The station-LOSO R^2 of 0.775 is substantially higher than the MLR and GTWR baselines. The bootstrap interval and paired station tests make this result more credible than a single aggregate point estimate because they account for station-level heterogeneity.

The validation design is central to the finding. Random K-fold performance is reported, but it is not the headline result because repeated records from the same station can appear in both training and test sets. Station-LOSO is more demanding and better matches the practical question: can the model predict a location where no station has been used for training? The answer is positive for the current station network, but this conclusion should not be stretched into unvalidated claims about every road segment.

The results also explain why the RMSE remains high in absolute terms. Delhi PM2.5 has large seasonal and episodic peaks, especially during postmonsoon and winter. In the all-features modeling table, mean 10 AM PM2.5 is 119.7 ug/m3 and the maximum reaches 598.8 ug/m3. The station-LOSO RMSE of 39.75 ug/m3 is therefore 33.2% of the mean. That absolute error is still high for local regulatory interpretation, but it is reasonable for city-scale prediction in a setting where AQI levels and seasonal concentration swings are very high. The seasonal table shows this clearly: postmonsoon has the strongest R^2 and the highest RMSE, while monsoon has lower RMSE but lower R^2 because the concentration range is compressed.

The comparison with GTWR is also informative. GTWR is attractive for spatial air-pollution modeling because it adapts coefficients locally, but the Delhi station network is still sparse for high-dimensional local regression. The LightGBM model appears to use nonlinear combinations of season, AOD, pressure, temperature, vegetation, reflectance, and OSM context more effectively than the local linear baselines. The Shadipur diagnostic is a reminder that GTWR can still outperform LightGBM at individual stations with unusual local context.

An external PM2.5 product benchmark pipeline has been implemented, but external product metrics are not yet reported because the station-date external product file is not present locally. CAMS is the most direct benchmark for the 2021-2025 study period. The Dey et al. India 1-km product is relevant but has a 2000-2019 period, so any comparison needs to state that mismatch clearly.

VI. LIMITATIONS AND NEXT WORK

First, the current model is validated at stations, not on independent mobile or temporary-monitor measurements. This means the model is appropriate for station-level spatial inference and station-gap filling, but road-level surfaces still need separate validation. Second, the external PM2.5 benchmark still needs station-date product values before the paper can claim improvement over an off-the-shelf gridded product.

The next work is a road-based healthy route planning app. In that version, each road edge receives a predicted PM2.5 value from the Delhi exposure surface, and the routing engine compares the shortest path with a lower-exposure path using an exposure-weighted edge cost. The app should report both travel distance or time and estimated PM2.5 exposure so that users can see the trade-off between route length and cleaner travel.

VII. CONCLUSION

This study shows that Delhi PM2.5 can be predicted at unseen monitoring stations using satellite TOA reflectance, AOD, meteorology, terrain, and microscale geographic predictors. Under station-LOSO validation, LightGBM_d5 achieved R^2 = 0.775 and RMSE = 39.75 ug/m3. Station-cluster bootstrap intervals, paired station tests, residual spatial diagnostics, seasonal metrics, and buffer-radius sensitivity all support the main station-level result.

The work establishes a rigorous station-validated base for Delhi exposure assessment. The next phase is implementation, not a new modeling claim: extend the same feature schema to road or grid prediction points, clip surfaces to official boundaries, generate seasonal maps, benchmark against external PM2.5 products, and validate dense surfaces with independent observations when they become available.

REFERENCES

[1] S. Dey et al., "A satellite-based high-resolution (1-km) ambient PM2.5 database for India over two decades (2000-2019)," *Remote Sensing*, vol. 12, no. 23, Art. no. 3872, 2020, doi: 10.3390/rs12233872.

[2] Copernicus Atmosphere Monitoring Service, CAMS global reanalysis and atmospheric composition products. Full product citation to be added after selecting the benchmark product.

[3] Central Pollution Control Board, India, continuous ambient air quality monitoring data. Full data access citation to be added.

[4] U.S. Geological Survey, Landsat 8 data products. Full collection citation to be added.

[5] OpenStreetMap contributors, OpenStreetMap geographic data. Full citation and access date to be added.

[6] C. Tong, Z. Shi, W. Shi, and A. Zhang, "Estimation of on-road PM2.5 distributions by combining satellite top-of-atmosphere with microscale geographic predictors for healthy route planning," *GeoHealth*, vol. 6, no. 9, e2022GH000669, 2022, doi: 10.1029/2022GH000669.

APPENDIX TABLE

TABLE A1
STATION INVENTORY USED IN THE DELHI MODELING TABLE

Station ID	CPCB station	Agency	Lat	Lon	Records	Missing %	Type
CPCB_10110	Dr. Karni Singh Shooting Range, Delhi	DPCC	28.4997	77.2671	1780	2.52	unknown
CPCB_10111	Major Dhyan Chand National Stadium, Delhi	DPCC	28.6125	77.2374	1769	3.12	unknown
CPCB_10112	Nehru Nagar, Delhi	DPCC	28.5668	77.2514	1772	2.96	unknown
CPCB_10113	Jahangirpuri, Delhi	DPCC	28.7330	77.1720	1740	4.71	unknown
CPCB_10114	Wazirpur, Delhi	DPCC	28.7005	77.1656	1744	4.49	unknown
CPCB_10115	Ashok Vihar, Delhi	DPCC	28.6957	77.1813	1735	4.98	unknown
CPCB_10116	Okhla Phase-2, Delhi	DPCC	28.5313	77.2707	1786	2.19	unknown
CPCB_10117	Rohini, Delhi	DPCC	28.7327	77.1188	1788	2.08	unknown
CPCB_10118	Vivek Vihar, Delhi	DPCC	28.6721	77.3138	1722	5.70	unknown
CPCB_10119	Dwarka-Sector 8, Delhi	DPCC	28.5769	77.0759	1801	1.37	unknown
CPCB_10120	Najafgarh, Delhi	DPCC	28.5727	76.9334	1721	5.75	unknown
CPCB_10121	Sonia Vihar, Delhi	DPCC	28.7101	77.2462	1756	3.83	unknown
CPCB_10122	Lodhi Road, Delhi	IITM	28.5897	77.2219	1105	39.49	unknown
CPCB_10123	CRRI Mathura Road, Delhi	IMD	28.5517	77.2750	1704	6.68	unknown
CPCB_10124	Pusa, Delhi	DPCC	28.6370	77.1722	1750	4.16	unknown
CPCB_10125	Burari Crossing, Delhi	IMD	28.7256	77.2034	1322	27.60	unknown
CPCB_10126	Aya Nagar, Delhi	IMD	28.4829	77.1267	1738	4.82	unknown
CPCB_10704	Patparganj, Delhi	DPCC	28.6202	77.2877	1755	3.89	unknown
CPCB_10705	Jawaharlal Nehru Stadium, Delhi	DPCC	28.5828	77.2344	1745	4.44	unknown
CPCB_10706	Narela, Delhi	DPCC	28.8206	77.1011	1780	2.52	unknown
CPCB_10707	Sri Aurobindo Marg, Delhi	DPCC	28.5283	77.1893	1766	3.29	unknown
CPCB_10708	Mundka, Delhi	DPCC	28.6824	77.0305	1751	4.11	unknown
CPCB_11266	Alipur, Delhi	DPCC	28.8157	77.1525	1753	4.00	unknown
CPCB_11267	Bawana, Delhi	DPCC	28.7758	77.0463	1783	2.35	unknown
CPCB_2553	Anand Vihar, Delhi	DPCC	28.6508	77.3152	1671	8.49	unknown
CPCB_2554	Mandir Marg, Delhi	DPCC	28.6341	77.2005	1674	8.32	unknown
CPCB_2555	Punjabi Bagh, Delhi	DPCC	28.6683	77.1167	1735	4.98	unknown
CPCB_2556	R K Puram, Delhi	DPCC	28.5648	77.1744	1747	4.33	unknown
CPCB_3715	ITO, Delhi	CPCB	28.6286	77.2410	1691	7.39	unknown
CPCB_8179	Shadipur, Delhi	CPCB	28.6516	77.1580	1738	4.82	unknown
CPCB_9290	DTU, Delhi	CPCB	28.7497	77.1163	1635	10.46	unknown